

CoLOR-DP: Conjugate Low-Rank Differential Privacy for Structure-Aware LoRA Fine-Tuning

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Abstract

Web developers increasingly fine-tune self-hosted Large Language Models (LLMs) with Low-Rank Adaptation (LoRA) to power intelligent web services. However, this practice introduces significant privacy risks, as sensitive user data can be memorized and exposed. While DP-SGD offers formal privacy guarantees, its isotropic noise neglects the low-rank structure of LoRA updates. This mismatch causes substantial utility loss on LLMs, forcing developers to abandon self-hosted models in favor of costly third-party APIs. In this paper, we propose Conjugate Low-Rank Differential Privacy (CoLOR-DP), a structure-aware mechanism that aligns DP with LoRA's inherent geometry. CoLOR-DP first projects clipped gradients onto a rank- r' subspace to reduce sensitivity, then injects conjugate Gaussian noise with negative correlation between LoRA factors. This design lowers the effective noise variance by a factor of $\frac{r'}{r}(1-\rho)$ while preserving (ϵ, δ) -DP guarantees. Across diverse benchmarks and LLM architectures, CoLOR-DP consistently outperforms DP-SGD under the same privacy budgets, achieving higher accuracy, lower overhead, and a practical solution to deploying private and self-hosted web services.

CCS Concepts

• Security and privacy → Privacy-preserving protocols.

Keywords

Differential Privacy, Large Language Models, LoRA

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1 Introduction

The modern web is witnessing a paradigm shift toward self-hosted intelligence. Increasingly, services that power search, recommendations, conversational assistance, and content moderation are embedding self-hosted large language models (LLMs)¹ to embed model-driven features while retaining control over data and infrastructure [5, 10, 19, 32, 36]. To achieve low latency, cost efficiency, and domain-specific customization, many teams fine-tune these models with Low-Rank Adaptation (LoRA) [3, 27] on proprietary data [7, 12, 17]. However, this practice introduces significant privacy and security concerns: fine-tuned models may inadvertently memorize and expose sensitive information through shared models, public endpoints, or CI/CD artifacts. Meanwhile, regulations such as GDPR [11] and CCPA [25] require privacy by design, creating a pressing challenge: How to retain the advantages of self-hosted LLMs while ensuring robust privacy protection during fine-tuning.

Differential Privacy (DP) [2, 34] provides a mathematically rigorous framework for bounding information leakage during training. The standard approach, DP-SGD [1, 8, 16, 22], clips per-sample gradients and injects calibrated Gaussian noise to satisfy (ϵ, δ) -DP. However, DP-SGD is designed for full-rank gradient updates in deep learning and treats all dimensions equally through isotropic noise. This design fundamentally mismatches the low-rank structure of LoRA fine-tuning. In LoRA, updates decompose as BA where

¹See <https://techcrunch.com/2025/05/31/google-quietly-released-an-app-that-lets-you-download-and-run-ai-models-locally> and <https://www.infoq.com/news/2025/03/gitlab-self-hosted-ai-platforms/>

$B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with $r \ll \min(d, k)$. The useful signal concentrates in a low-dimensional subspace, yet DP-SGD injects noise uniformly across all r dimensions. This mismatch leads to severe utility degradation, especially on capacity limited LLMs where representation power is constrained. When accuracy drops substantially, web developers abandon self-hosted models in favor of costly third-party APIs, defeating the purpose of local deployment.

Prior DP methods address parts of this challenge but leave critical gaps, especially under web deployment constraints. Projection-based approaches [6] reduce dimensionality but do not account for LoRA factor coupling. Low-rank reparameterization [31] shrinks the update space yet perturbs factors independently, amplifying composed variance. PEFT-oriented methods [9, 30] and dynamic-rank schemes [28] improve over DP-SGD but still incur noticeable accuracy loss on LLMs under tight privacy budgets. These limitations compound in real-world web deployments, where self-hosted services operate on modest hardware with small batch sizes. It will amplify noise impact, making the trade-off between privacy and utility unfavorable. Web teams also require deterministic training for reproducible CI/CD pipelines and fixed iteration counts for stable privacy accounting. The core gap remains: no existing method simultaneously preserves LoRA's representation capacity, reduces noise impact through structured perturbation, and integrates seamlessly into resource-constrained web pipelines. Our work attempts to bridge this gap at the web-application layer, aligning privacy-preserving LoRA fine-tuning with deployment demands such as low latency, endpoint safety, and reproducible workflows.

In order to address these issues, we present a Conjugate Low-Rank Differential Privacy (CoLoR-DP) mechanism for LoRA-based LLMs fine-tuning in web deployments. It keeps the model and toolchain unchanged and relies on fixed training steps for stable accounting. CoLoR-DP has two steps. First, it projects clipped per-sample gradients onto a rank- r' subspace that captures the dominant update energy. This concentrates the learning signal and reduces the effective sensitivity seen by the DP mechanism. The projection basis is updated at a low frequency to limit overhead. Second, it injects conjugate Gaussian noise into the LoRA factors A and B within that subspace. The two noise vectors are negatively correlated with a coefficient ρ . When the factors compose into BA , the dominant noise terms partially cancel. The resulting variance on the composed update is reduced by a factor of $\frac{r'}{r}(1 - \rho)$ while maintaining (ϵ, δ) -DP guarantees. In general, CoLoR-DP adds minimal overhead and fits the cost and latency budgets of self-hosted, LLM-powered web services.

We evaluate CoLoR-DP in scenarios that emulate resource-constrained web deployments. Experiments use LLMs ranging from 1.5B to 7B parameters across multiple models, fine-tuned with LoRA adapters via standard toolchains on benchmark tasks [23]. We test privacy budgets $\epsilon \in \{1, 5, 8\}$ with $\delta = 10^{-6}$ to cover both stringent and moderate protection levels typical in web applications. Baselines include DP-SGD under identical privacy accounting. Beyond standard utility metrics (task accuracy and training loss), we validate privacy guarantees empirically by measuring resistance to SOTA membership inference attacks (MIAs) in both black-box and white-box settings. Results demonstrate that CoLoR-DP consistently outperforms baselines across privacy budgets, achieving up

to 10.44% accuracy improvement at $\epsilon = 1$ while maintaining provable privacy and practical efficiency for self-hosted web services.

- We propose CoLoR-DP, a novel DP mechanism that exploits LoRA's low-rank structure through two synergistic techniques: (i) gradient projection onto a rank- r' subspace to concentrate learning signal and reduce effective sensitivity, and (ii) conjugate noise injection with correlation coefficient ρ between LoRA factors A and B to achieve noise cancellation in the composed update BA .
- We provide rigorous privacy analysis proving that CoLoR-DP satisfies (ϵ, δ) -DP while reducing noise variance by a factor of $\frac{r'}{r}(1 - \rho)$ compared to standard DP-SGD. This theoretical result formalizes the privacy-utility advantage of exploiting conjugate structure in low-rank updates.
- We demonstrate through extensive experiments on GLUE benchmarks with LLMs (1.5B–7B parameters) that CoLoR-DP achieves 1.5%–10.44% accuracy improvements over DP-SGD across privacy budgets $\epsilon \in \{1, 5, 8\}$, with empirical privacy validated via MIA.
- We show that CoLoR-DP integrates seamlessly into standard LoRA toolchains, making privacy-preserving fine-tuning practical for resource-constrained web deployments.

2 Preliminaries and Problem Formalization

We introduce LoRA parameterization and DP definitions, then analyze structural challenges when applying DP-SGD to LoRA fine-tuning in web deployments.

2.1 LoRA Fine-Tuning and Differential Privacy

2.1.1 LoRA Parameterization. Web services increasingly fine-tune the self-hosted LLMs to enhance task specific performance while controlling costs. Full fine-tuning of LLMs is prohibitive on modest GPUs, making parameter efficient methods essential. LoRA [4, 20, 26] addresses this by freezing the pre-trained backbone and injecting trainable low-rank matrices into linear layers.

Definition 2.1 (LoRA Parameterization). For a linear layer with weight matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, LoRA introduces two trainable matrices $A \in \mathbb{R}^{r \times d_{\text{in}}}$, and $B \in \mathbb{R}^{d_{\text{out}} \times r}$, for $r \ll d_{\text{in}}, d_{\text{out}}$. And defines the effective weight as

$$W = W_0 + \frac{\alpha}{r} BA, \quad (1)$$

where r is the LoRA rank and α is a scaling factor.

This reduces trainable parameters from $d_{\text{in}}d_{\text{out}}$ to $r(d_{\text{in}} + d_{\text{out}})$, enabling fine-tuning on resource-constrained hardware typical of web deployments. The gradient structure follows from chain rule: $\nabla_A L = \frac{\alpha}{r} B^T G$ and $\nabla_B L = \frac{\alpha}{r} G A^T$, where $G = \nabla_W L$ (as detailed in Appendix A.1). Table 1 summarizes our used notation.

2.1.2 Differential Privacy. Web deployments handle sensitive data (such as user logs, customer tickets, and proprietary documents), raising concrete privacy risks. Regulations like GDPR mandate privacy by design, while model memorization and MIAs pose technical threats. DP offers a rigorous way to bound information leakage.

Definition 2.2 ((ϵ, δ)-DP). A randomized mechanism $\mathcal{M} : \mathcal{D} \rightarrow \mathcal{R}$ satisfies (ϵ, δ) -DP [2, 33, 35] if for any adjacent datasets D and D'

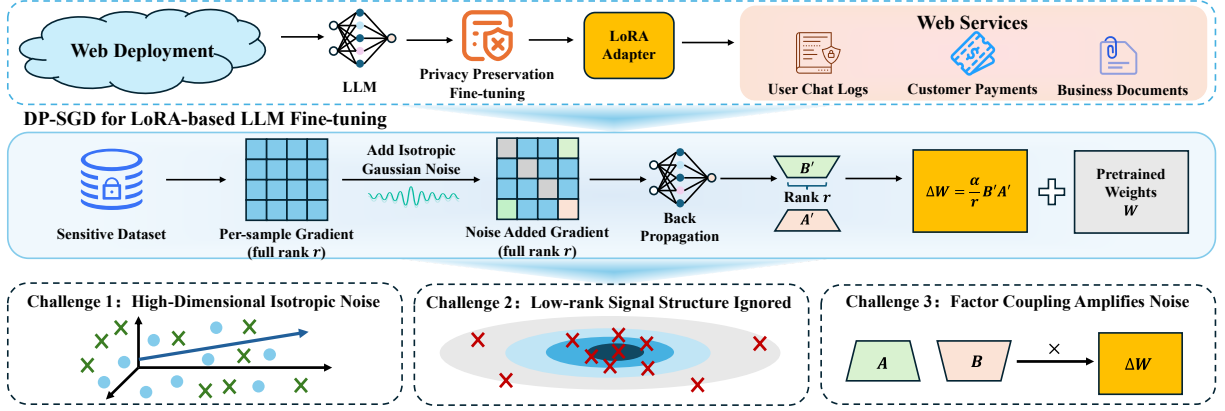


Figure 1: The research challenges of applying DP-SGD to LoRA-based LLM Fine-tuning

Table 1: Notation used in the CoLoR-DP mechanism.

Symbol	Description
d_{in}, d_{out}	Input/output dimensions of LoRA modules
r	Rank of LoRA matrices A and B
$A \in \mathbb{R}^{r \times d_{in}}, B \in \mathbb{R}^{d_{out} \times r}$	LoRA trainable parameters
$r' \leq r$	Projection rank in low-rank subspace
ϵ, δ	DP parameters
σ	Noise multiplier for Gaussian mechanism
C_0	Original gradient clipping threshold
$\kappa(r')$	Energy retention ratio after projection
$C' = \sqrt{\kappa(r')} C_0$	Adjusted clipping after projection
$\rho \in [0, 1]$	Correlation coefficient for conjugate noise
q	Poisson sampling rate in SGD
T	Total number of training iterations

differing in one record and for any measurable set $S \subseteq \text{Range}(\mathcal{M})$:

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta, \quad (2)$$

where ϵ indicates privacy strength (smaller is stronger), while δ allows negligible failure probability.

Iterative training requires composing privacy guarantees across T steps. Rényi DP (RDP) [13] enables tighter accounting than naive composition (as detailed in Appendix A.2).

2.2 DP-SGD on LoRA: Baseline and Challenges

To apply DP to LoRA fine-tuning, we adopt DP-SGD [1, 30], which clips per-sample gradients and injects calibrated noise. At iteration t , a minibatch $\mathcal{B}_t$ is sampled via Poisson subsampling with rate $q = |\mathcal{B}_t|/N$. For each sample $i \in \mathcal{B}_t$, the LoRA gradients $\nabla_A L_i$ and $\nabla_B L_i$ are computed and concatenated.

Definition 2.3 (DP-SGD for LoRA [30]). The per-sample gradient vector is

$$g_i = [\nabla_A L_i; \nabla_B L_i] \in \mathbb{R}^m, \quad m = r(d_{in} + d_{out}), \quad (3)$$

which is clipped to ℓ_2 -norm C_0 as

$$\tilde{g}_i = g_i / \max(1, \|g_i\|_2 / C_0), \quad (4)$$

and the batch update is perturbed with Gaussian noise as follows

$$\tilde{g} = \frac{1}{|\mathcal{B}_t|} \sum_{i \in \mathcal{B}_t} \tilde{g}_i + \mathcal{N}(0, \sigma^2 C_0^2 I_m). \quad (5)$$

This mechanism satisfies (ϵ, δ) -DP via RDP accounting. However, applying DP-SGD directly to LoRA reveals three structural challenges that compound utility loss in resource-constrained web deployments, as illustrated in Figure 1.

Challenge #1: High-dimensional isotropic noise. The noise covariance $\sigma^2 C_0^2 I_m$ injects equal perturbation across all m dimensions. This uniform treatment ignores the fact that LoRA gradients concentrate energy in a much lower-dimensional subspace, thereby wasting privacy budget on irrelevant directions. Small batch sizes typical in web deployments ($q \approx 0.01$) amplify this inefficiency, as fewer samples average out the noise, leading to slow convergence and substantial accuracy loss on capacity limited LLMs.

Challenge #2: Low-rank concentration of gradient energy. Empirically, most gradient information lies in a subspace of rank $r' \ll m$. This mirrors observations in deep learning [6, 31] but is amplified in LoRA due to explicit low-rank constraints. Standard DP-SGD perturbs all directions equally, leading to inefficient privacy budget allocation. The mismatch is particularly severe when $r'/m \ll 1$, as the mechanism overprotects dimensions that contribute negligibly to model updates.

Challenge #3: Multiplicative noise amplification via factor coupling. LoRA updates compose as $\Delta W = BA$. When DP-SGD adds independent noise Z_A, Z_B to the factors, the composed update becomes as $(B + Z_B)(A + Z_A) = BA + BZ_A + Z_B A + Z_B Z_A$. The cross terms amplify effective noise variance beyond independent perturbation, unique to low-rank factorizations.

These three challenges motivate the design of our CoLoR-DP mechanism. In Section 3, we address all three challenges through low-rank gradient projection (Challenges #1 and #2) and conjugate noise injection (Challenge #3).

3 The Proposed CoLoR-DP Mechanism

Standard DP-SGD applies isotropic noise to LoRA gradients without considering their inherent low-rank structure, leading to unnecessary utility loss. We propose CoLoR-DP, which projects clipped

gradients onto reduced subspaces and applies conjugate Gaussian noise within these dimensions. This section presents the mechanism design, variance reduction analysis, and privacy guarantees.

3.1 Synergistic Design: Projection and Conjugate Noise

CoLoR-DP integrates two complementary components. First, *low-rank projection* projects each clipped per-sample gradient onto an r' -dimensional subspace that captures dominant gradient energy (Section 3.2). Second, *conjugate Gaussian noise* injects negatively correlated noise vectors (z_A, z_B) controlled by correlation coefficient $\rho \in [0, 1]$ into the LoRA factors within that subspace (Section 3.3). Projection eliminates noise in low-information directions, while conjugate noise reduces variance amplification from independent factor perturbation. Algorithm 1 outlines the complete procedure. Corollary 3.8 formalizes the variance reduction guarantee.

3.2 Low-Rank Gradient Projection

We project clipped gradients onto a rank- r' subspace ($r' \ll m$ with $m = r(d_{\text{in}} + d_{\text{out}})$) to concentrate noise on meaningful directions. Given clipped per-sample gradients $\hat{g}_i = g_i \cdot \min(1, C_0/\|g_i\|_2)$ for $i = 1, \dots, B$, we construct the projection basis from batch statistics.

Definition 3.1 (Subspace Selection). Let $\{\hat{g}_i\}_{i=1}^B$ denote the clipped per-sample gradients. We compute the covariance matrix $\Sigma_g = \frac{1}{B} \sum_{i=1}^B \hat{g}_i \hat{g}_i^\top$ and perform SVD to obtain

$$\Sigma_g = U_{\text{full}} \Lambda U_{\text{full}}^\top, \quad (6)$$

where Λ contains singular values $\{\lambda_j\}$ in descending order. The projection matrix $U \in \mathbb{R}^{m \times r'}$ comprises the first r' columns of U_{full} .

To minimize overhead, we update U every T_{proj} iterations rather than per step, exploiting the relatively stable spectrum of Σ_g . The projected gradient is defined as follows.

Definition 3.2 (Low-Rank Gradient Projection). Let $P_{r'} = UU^\top$ denote the orthogonal projection operator onto the rank- r' subspace. For gradient $g \in \mathbb{R}^m$, the projected gradient is

$$g^{\text{proj}} = P_{r'} g = UU^\top g. \quad (7)$$

To quantify the proportion of gradient energy retained after projection, we define the energy retention ratio as follows.

Definition 3.3 (Energy Retention Ratio). Let $\{\lambda_j\}_{j=1}^m$ be the singular values of Σ_g . The energy retention ratio is

$$\kappa(r') = \frac{\sum_{j=1}^{r'} \lambda_j}{\sum_{j=1}^m \lambda_j} \in [0, 1]. \quad (8)$$

This measures the proportion of gradient variance preserved in the rank- r' subspace. Since clipping was performed in the full space with threshold C_0 , we adjust the threshold for the projected subspace.

Definition 3.4 (Energy-Aware Clipping Threshold). The clipping threshold for projected gradients is

$$C' = \sqrt{\kappa(r')} C_0, \quad (9)$$

which scales the baseline threshold C_0 by the factor $\sqrt{\kappa(r')}$ to ensure that clipping aligns with the most informative gradient directions.

Algorithm 1: CoLoR-DP update for one mini-batch

Input: Mini-batch B , clip norm C_0 , projection rank r' , conjugacy ρ , learning rate η
Output: Private optimizer step U

- 1 **foreach** $i \in B$ **do**
- 2 Compute full gradient g_i ;
- 3 Clip $\hat{g}_i \leftarrow g_i \cdot \min(1, C_0/\|g_i\|_2)$;
- 4 **end**
- 5 **if** $t \bmod T_{\text{proj}} = 0$ **then**
- 6 recompute projection basis U
- 7 **end**
- 8 Projected batch gradient: $g^{\text{proj}} \leftarrow P_{r'} \left(\frac{1}{|B|} \sum_i \hat{g}_i \right)$;
- 9 Energy ratio $\kappa(r') \leftarrow \frac{\sum_{j=1}^{r'} \lambda_j}{\sum_{j=1}^m \lambda_j}$, Adjusted clip $C' \leftarrow \sqrt{\kappa(r')} C_0$;
- 10 Draw $N, M \sim \mathcal{N}(0, I_{r'})$;
- 11 $z_A \leftarrow \sigma C' N$; $z_B \leftarrow -\rho \sigma C' N + \sqrt{1 - \rho^2} \sigma C' M$;
- 12 Noisy gradient: $g^{\text{noisy}} \leftarrow g^{\text{proj}} + [z_A^\top, z_B^\top]^\top$;
- 13 Optimizer step (e.g. AdamW): $U \leftarrow -\eta g^{\text{noisy}}$;
- 14 **add_step**(g, σ) to RDP accountant;
- 15 **return** U ;

REMARK 3.1 (ROLE OF THE PROJECTOR). The operator $P_{r'}$ removes gradient components orthogonal to the selected subspace. Once the projection basis U is fixed, $P_{r'}$ is data independent, and applying it constitutes post-processing that adds no extra privacy cost. In practice, U is periodically updated from clipped batch statistics, making each update data-dependent. We conservatively treat these updates as composed DP steps with bounded sensitivity, since U is a deterministic function of clipped gradients with $\|g_i\|_2 \leq C_0$, ensuring finite sensitivity of Σ_g and its subspace (see proof in Appendix B.3). Accounting for these updates under standard RDP composition yields a conservative overall (ϵ, δ) -DP guarantee. In Section 3.4, we further bound the ℓ_2 -sensitivity by $\|P_{r'} \hat{g}_i\|_2 \leq C'$, tightening the RDP analysis.

REMARK 3.2 (CHOOSING r'). The projection rank balances noise reduction against information retention. Smaller r' confines noise to fewer dimensions, improving efficiency but potentially discarding informative gradients. When $r' = r$, it recovers standard DP-SGD. We find $r' \in [0.5r, 0.75r]$ effective in practice, corresponding to $r' \in [4, 48]$ for typical ranks $r \in [8, 64]$.

3.3 Conjugate Gaussian Noise in the Projected Subspace

Low-rank projection does not address the coupling between A and B . As they act via BA , independent noise on each factor amplifies variance linearly with r' . We quantify this below.

LEMMA 3.5 (VARIANCE UNDER INDEPENDENT NOISE). Let $z_A, z_B \sim \mathcal{N}(0, \sigma^2 C'^2 I_{r'})$ be independent noise vectors in the projected subspace, and define $\Delta W = (B + z_B)(A + z_A)^\top - BA = Bz_A^\top + z_B A^\top + z_B z_A^\top$. Omitting the quadratic term $z_B z_A^\top$ (which is $\mathcal{O}(\sigma^4 C'^4)$, two orders smaller than the dominant mixed terms), the variance is

$$\text{Var}[\|\Delta W\|_F^2] = \sigma^2 C'^2 (\|A\|_F^2 + \|B\|_F^2) + r' \sigma^4 C'^4. \quad (10)$$

The detailed derivation of Lemma 3.5, including the expansion of all cross-terms and the justification for omitting higher-order terms, is provided in Appendix B.1.

To suppress the additional variance identified in Lemma 3.5, we inject negatively correlated Gaussian vectors.

Definition 3.6 (Conjugate Gaussian Noise). Let $z = [z_A^\top, z_B^\top]^\top \in \mathbb{R}^{2r'}$ satisfy

$$z \sim \mathcal{N}(0, \Sigma_\rho), \quad \Sigma_\rho = \sigma^2 C'^2 \begin{bmatrix} I_{r'} & -\rho I_{r'} \\ -\rho I_{r'} & I_{r'} \end{bmatrix}, \quad (11)$$

where $\rho \in [0, 1)$ controls the negative correlation. Equivalently, draw $N, M \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_{r'})$ and set

$$z_A = \sigma C' N, \quad z_B = -\rho \sigma C' N + \sqrt{1 - \rho^2} \sigma C' M. \quad (12)$$

To map the projected noise back to full dimension, we define the inverse mapping as follows.

Definition 3.7 (Inverse Mapping to Full Space). Let $U \in \mathbb{R}^{m \times r'}$ be the projection basis. Define the block matrix $U_\oplus = \text{diag}(U, U) \in \mathbb{R}^{2m \times 2r'}$. The full-dimensional noise is

$$z_{\text{full}} = U_\oplus z, \quad (13)$$

preserving dimensional consistency and correlation structure for both A and B updates.

By coupling the noise vectors, the conjugate mechanism achieves substantial variance reduction.

COROLLARY 3.8 (VARIANCE REDUCTION WITH CONJUGATE NOISE). Let z be drawn from Definition 3.6. Omitting the quadratic product as in Lemma 3.5, the variance becomes

$$\begin{aligned} \text{Var}[\| \Delta W \|_F^2] &= \sigma^2 C'^2 (\|A\|_F^2 + \|B\|_F^2) + (1 - \rho) r' \sigma^4 C'^4 \\ &= \sigma^2 \kappa(r') C_0^2 (\|A\|_F^2 + \|B\|_F^2) + \boxed{\frac{r'}{r} (1 - \rho) \sigma^4 C_0^4}, \end{aligned} \quad (14)$$

where the second line substitutes $C'^2 = \kappa(r') C_0^2$ with $\kappa(r') \approx r'/r$. The boxed term is the leading noise contribution after projection and conjugate coupling. The factor $\frac{r'}{r} (1 - \rho)$ quantifies variance reduction compared to standard DP-SGD.

The complete proof of Corollary 3.8, including the calculation of all variance terms under the correlated noise structure, is provided in Appendix B.1.

REMARK 3.3 (CHOOSING ρ). A larger ρ reduces noise amplification in BA and improves accuracy. Empirically, $\rho \in [0.7, 0.95]$ works well for LoRA ranks $r \in [8, 64]$.

3.4 Privacy Analysis

We prove that CoLoR-DP satisfies differential privacy using RDP accounting. Our analysis establishes RDP bounds for each step and applies composition to achieve the overall (ϵ, δ) -DP guarantee.

3.4.1 Single-Step Privacy Analysis. Each CoLoR-DP iteration computes the mechanism output

$$\mathcal{M}_{\text{CoLoR}}(D) = \frac{1}{B} \sum_{i=1}^B (P_{r'} \hat{g}_i + U [z_A \ z_B]^\top), \quad (15)$$

where $P_{r'}$ is the orthogonal projection, $\hat{g}_i$ the clipped gradient, and (z_A, z_B) the conjugate Gaussian noise. We first bound the sensitivity.

COROLLARY 3.9 (SENSITIVITY BOUND). For adjacent datasets D and D' differing in one record, the ℓ_2 -sensitivity of the projected gradient is bounded by

$$\|\mu - \mu'\|_2 = \left\| \frac{1}{B} (P_{r'} \hat{g}_i - P_{r'} \hat{g}'_i) \right\|_2 \leq \frac{C'}{B} = \frac{\sqrt{\kappa(r')} C_0}{B}. \quad (16)$$

The conjugate noise covariance Σ_ρ has minimum eigenvalue $\lambda_{\min}(\Sigma_\rho) = \sigma^2 C'^2 (1 - \rho)/B^2$. Applying the Gaussian RDP mechanism, we obtain the privacy guarantee for a single training step.

THEOREM 3.10 (SINGLE-STEP RDP). For any $\alpha > 1$, each CoLoR-DP iteration satisfies $(\alpha, \epsilon_\alpha^{\text{step}})$ -RDP with

$$\epsilon_\alpha^{\text{step}} = \frac{\alpha}{2\sigma^2(1 - \rho)}. \quad (17)$$

The proof of Theorem 3.10 follows from standard Gaussian RDP analysis [13] applied to our sensitivity bound and noise covariance. The detailed calculation is provided in Appendix B.2.

3.4.2 Overall Privacy Guarantee. With Poisson subsampling at rate $q = B/N$ and T training iterations, we apply standard RDP composition and privacy amplification via subsampling [24]. The combined effect yields the following guarantee.

COROLLARY 3.11 (OVERALL DP GUARANTEE). After T iterations with Poisson subsampling at rate q , CoLoR-DP ensures (ϵ, δ) -DP with

$$\epsilon = \min_{\alpha > 1} \left\{ T \cdot \epsilon_\alpha^{\text{amp}} + \frac{\log(1/\delta)}{\alpha - 1} \right\}, \quad (18)$$

where $\epsilon_\alpha^{\text{amp}} = \frac{1}{\alpha - 1} \log(1 + q^2 \alpha (\alpha - 1) e^{(\alpha - 1) \epsilon_\alpha^{\text{step}}})$ accounts for privacy amplification.

The derivation of Corollary 3.11 involves applying the subsampled RDP composition lemma and converting from RDP to (ϵ, δ) -DP. These are detailed in Appendix B.2.

REMARK 3.4 (IMPACT OF CoLoR-DP COMPONENTS). The projection $P_{r'}$ and reconstruction U are post-processing operations that preserve privacy. The conjugate noise effectively reduces variance along the least-protected direction by a factor $(1 - \rho)$, improving utility while maintaining the privacy guarantee in Theorem 3.10.

3.5 Implementation of CoLoR-DP

The CoLoR-DP training pipeline extends standard LoRA fine-tuning with three components: a projection basis manager, a conjugate noise sampler, and a privacy accountant. Before training begins, we use the Opacus RDP accountant to determine the minimal noise multiplier σ^* satisfying the privacy budget (ϵ, δ) for a given dataset size N , batch size B , and epochs E . This involves binary search over candidate σ values until the privacy constraint is met. The complete accounting algorithm is provided in Appendix B.4. Once σ^* is fixed, training runs for exactly $T = E \lceil N/B \rceil$ iterations without early stopping to maintain deterministic privacy accounting.

Training Process. As demonstrated in Figure 2, each iteration processes mini-batches through the following steps:

- (1) Parameter Initialization.** Compute the noise multiplier σ^* using the privacy accounting procedure (Appendix B.4). This value remains fixed throughout training.

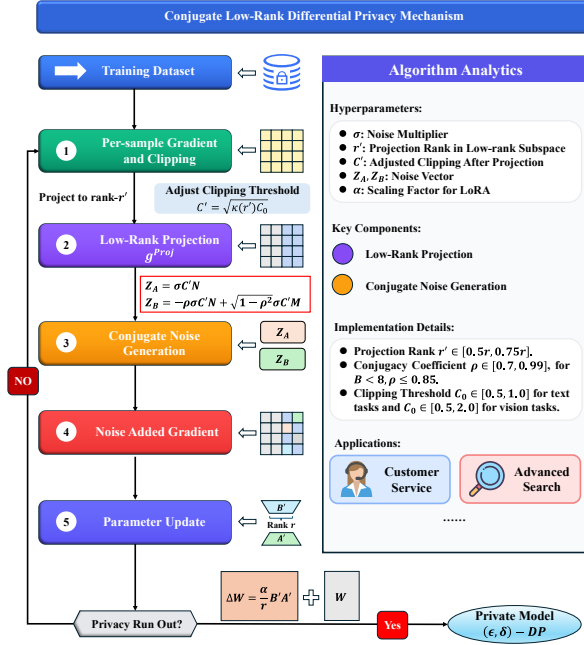


Figure 2: Overall workflow of the CoLOR-DP mechanism.

- (2) **Per-Sample Gradient and Clipping.** Compute per-sample gradients for LoRA parameters (A, B) and clip each to ℓ_2 -norm C_0 .
- (3) **Low-Rank Projection.** Update the projection basis $P_{r'}$ every T_{proj} iterations using SVD on gradient covariance statistics. Between updates, project clipped gradients onto the rank- r' subspace: $g^{\text{proj}} = P_{r'} \hat{g} = U U^T \hat{g}$.
- (4) **Conjugate Noise Generation.** Sample correlated Gaussian vectors $(z_A, z_B) \in \mathbb{R}^{r'}$ with variance $\sigma^2 C'^2$ and correlation coefficient ρ . Map to full dimension via $z_{\text{full}} = U \oplus z$.
- (5) **Noise Added Gradient.** Add the conjugate noise to the projected gradient: $\tilde{g}^{\text{proj}} = g^{\text{proj}} + z_{\text{full}}$.
- (6) **Parameter Update.** Average noisy gradients across the mini-batch and update (A, B) via standard optimizers while keeping pre-trained backbone weights frozen.

Implementation Notes. Operating exclusively on LoRA parameters, CoLOR-DP requires only $O(r'd_{\text{in}} + r'd_{\text{out}})$ additional memory and integrates into standard training loops without architectural changes. The periodic SVD (every $T_{\text{proj}} = 300$ steps by default) only adds less than 3% runtime overhead compared to DP-SGD.

4 Experiments

In this section, we present the experimental setup and results.

4.1 Experimental Setup

4.1.1 Datasets. We evaluate on six text classification benchmarks from Hugging Face. Four come from the GLUE benchmark [23]: SST-2 (67K/872 samples for training/validation), QNLI (105K/5.5K, question-answering entailment), MNLI (393K/9.8K, natural language inference), and QQP (364K/40K, semantic similarity). We

additionally use AG News² (120K/7.6K, 4-class news categorization) and IMDB³ (25K/25K, binary movie review sentiment). All experiments use predefined splits without additional preprocessing and report accuracy on validation/test sets.

4.1.2 Backbone Models. We evaluate eight distilled model checkpoints spanning four prominent open source families: DeepSeek, Gemma, Llama, and Qwen. These models range from 1.5B to 7B parameters, matching the resource constraints of self-hosted web deployments. Despite their compact size, LLMs demonstrate strong performance on sentence-level classification tasks, which primarily require lexical understanding rather than complex long range reasoning. This parameter range represents the practical sweet spot for web services, balancing cost, latency, and utility.

We compare three methods: *Non-Private*, *DP-SGD*, and *CoLOR-DP*. The experimental setup and hyperparameter configuration are demonstrated in Appendix C.3.

4.2 Experimental Results

4.2.1 Accuracy Performance on Classification Tasks. Table 2 presents accuracy results for eight backbone models across four GLUE benchmarks (SST-2, QNLI, MNLI, QQP) under privacy budgets $\epsilon \in \{1, 5, 8\}$. CoLOR-DP consistently outperforms DP-SGD across all 96 configurations, with gains most pronounced under stringent constraints. At $\epsilon = 1$, CoLOR-DP achieves 1.5-3% higher accuracy on average, recovering over 90% of non-private performance. This gap narrows to 0.5-1.5% at $\epsilon = 8$, but CoLOR-DP maintains advantages across all privacy regimes. Notably, DeepSeek-R1-7B on SST-2 with CoLOR-DP at $\epsilon = 1$ (90.94%) approaches DP-SGD's performance at $\epsilon = 5$ (89.91%), achieving comparable utility with 5 $\times$ stronger privacy.

Model size significantly affects improvement magnitude. Smaller models (1-3B parameters) benefit most from structured noise injection, achieving 2-3% gains at $\epsilon = 1$. This is critical for web services where modest GPUs (e.g., RTX 4090) typically host such models for latency-sensitive applications. Gemma-3-1B improves from 91.97% to 92.89% on SST-2 at $\epsilon = 1$, translating to fewer production errors. Larger models like Qwen3-4B show smaller but consistent 1-2% gains. As privacy budgets tighten, CoLOR-DP's advantage increases: at $\epsilon = 1$, improvements reach 1-3%, while at $\epsilon = 8$, they narrow to 0.5-1%, aligning with our analysis (Corollary 3.8) that variance reduction becomes critical when noise dominates.

CoLOR-DP consistently matches or outperforms DP-SGD across all tasks. For self-hosted LLMs under strict privacy budgets ($\epsilon \leq 3$), the 1-3% gains make privacy-preserving fine-tuning practical while satisfying GDPR-style constraints.

4.2.2 Empirical Privacy via Membership Inference Attacks. Beyond theoretical (ϵ, δ) -DP guarantees, we validate empirical privacy using SOTA MIAs that mirror real-world threats to web-deployed models. Tables 3 and 4 report attack accuracy on AG News and IMDB datasets across $\epsilon \in \{1, 2, 3, 4\}$ for representative models under both black-box and white-box adversarial settings.

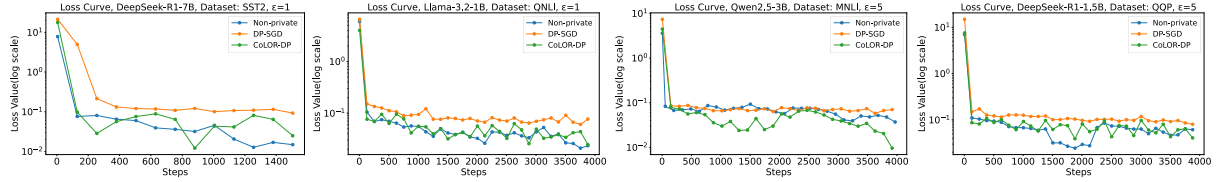
Black-Box/Shadow Attacks. This attack type, where adversaries train shadow models to mimic the target without accessing internal parameters, represents a realistic threat to web services exposing

²https://huggingface.co/datasets/fancyzhx/ag_news

³<https://huggingface.co/datasets/stanfordnlp/imdb>

Table 2: Accuracy results across different models and privacy budgets ($\epsilon = 1, 5, 8$) on SST2, QNLI, MNLI and QQP datasets.

Dataset	Method	ϵ	DeepSeek-R1-1.5B	DeepSeek-R1-7B	Gemma-3-1B	Llama-3.2-1B	Llama-3.2-3B	Qwen3-1.7B	Qwen3-4B	Qwen2.5-3B
SST2	Non-Private	N/A	92.43	94.94	94.15	95.07	94.79	95.09	95.15	94.99
	CoLOR-DP	1	91.86	90.94	92.89	92.23	93.30	93.81	94.99	93.53
		5	92.20	92.43	93.45	92.75	94.07	94.50	95.10	94.19
		8	92.78	93.35	93.72	93.35	94.30	94.61	95.33	94.90
	DPSGD	1	88.88	88.99	91.97	88.53	84.86	83.37	84.63	92.50
		5	90.48	89.91	93.46	90.48	88.99	86.47	89.68	93.50
QNLI	Non-Private	N/A	92.15	93.48	90.92	92.84	93.63	93.04	95.35	95.06
	CoLOR-DP	1	90.16	92.01	84.20	86.89	88.98	90.15	94.51	94.58
		5	91.43	92.66	85.58	88.16	90.52	91.84	94.78	94.62
		8	91.67	92.89	87.51	89.11	91.33	92.50	95.04	94.71
	DPSGD	1	89.93	90.79	83.31	86.31	88.65	86.95	93.19	89.46
		5	90.14	91.79	85.47	87.75	90.44	87.43	93.81	90.72
MNLI	Non-Private	N/A	87.05	89.68	85.01	85.91	90.31	91.01	91.19	91.25
	CoLOR-DP	1	85.52	86.04	84.06	84.08	86.89	84.86	89.95	89.89
		5	86.12	88.74	84.70	85.28	87.61	85.55	90.22	90.14
		8	86.84	88.82	84.95	85.45	87.87	85.70	90.12	90.89
	DPSGD	1	85.11	86.01	81.56	81.74	85.73	84.39	88.76	87.96
		5	86.15	88.20	83.10	83.93	86.91	85.21	89.42	88.51
QQP	Non-Private	N/A	91.67	91.27	87.91	89.99	91.56	88.33	90.20	89.89
	CoLOR-DP	1	88.02	89.49	85.92	85.94	87.07	84.95	87.49	87.72
		5	89.22	89.72	86.74	86.68	87.81	85.61	88.91	88.58
		8	89.28	89.84	86.81	86.98	87.96	85.82	89.84	89.53
	DPSGD	1	87.24	88.46	83.76	83.80	85.82	85.06	86.74	86.76
		5	88.35	89.16	85.42	85.20	86.51	85.84	87.29	87.28
		8	88.61	89.40	85.64	85.38	86.90	85.93	87.37	87.51

**Figure 3: Loss curves among different models and datasets.**

only prediction APIs. Both datasets show that CoLOR-DP and DP-SGD achieve attack accuracy near random guessing (0.50). On AG News at $\epsilon = 1$, CoLOR-DP yields 0.5021 and 0.5156 for DeepSeek-R1-1.5B and Llama-3.2-1B, respectively. Only 0-1.5% above the random baseline and 8-10% below non-private models (0.58-0.60). DP-SGD performs comparably (0.5064 and 0.5167), confirming strong protection against external probing. IMDb results mirror this pattern: Qwen3-1B achieves 0.5058 with CoLOR-DP versus 0.5158 with DP-SGD at $\epsilon = 1$, both substantially below the non-private baseline.

As privacy budgets relax to $\epsilon = 4$, attack accuracy increases slightly to 0.52-0.53 but remains far below non-private levels, demonstrating robust protection even under moderate privacy constraints. The consistency across datasets validates that CoLOR-DP’s privacy protection generalizes across task types typical of web applications. **White-Box/Partial Attacks.** This stronger adversary with access to gradients or partial model parameters represents insider threats or compromised deployment pipelines, relevant scenarios for web services with CI/CD artifacts or shared development environments. Attack success rates increase as expected, but CoLOR-DP maintains substantial protection. On AG News at $\epsilon = 1$, attack accuracy ranges from 0.5331 to 0.5601, approximately 3-6% above random guessing but 6-9% below non-private baselines (0.62-0.65). Notably, CoLOR-DP often demonstrates slightly better resistance than DP-SGD: on

DeepSeek-R1-1.5B, CoLOR-DP achieves 0.5331 versus DP-SGD’s 0.5475, a 1.4% advantage. IMDb results corroborate this pattern across both models tested, with CoLOR-DP consistently matching or outperforming DP-SGD. This suggests that conjugate noise’s correlation structure does not introduce exploitable patterns for gradient based attacks.

Cross referencing MIA results with accuracy confirms a favorable trade-off: CoLOR-DP achieves 2-3% higher accuracy than DP-SGD while maintaining comparable empirical privacy. Attack success rates within 2-7% of random guessing across all budgets validate that theoretical guarantees hold against practical adversaries.

4.2.3 Training Dynamics and Practical Insights. Figure 3 visualizes training loss curves across four datasets (*i.e.*, SST-2, QNLI, MNLI, QQP) for representative models. CoLOR-DP shows significantly improved convergence compared to DP-SGD. On SST-2 with DeepSeek-R1-1.5B, CoLOR-DP’s loss trajectory closely tracks the non-private baseline, converging to near identical final values. In contrast, DP-SGD exhibits slower convergence and higher plateaus, particularly on larger models such as DeepSeek-R1-7B, where the DP-SGD loss remains above 0.1 throughout training. Similar trends hold across datasets: CoLOR-DP achieves faster initial descent and more stable convergence, especially under tight privacy budgets

Table 3: Accuracy of MIAs on AGNews dataset.

Attack	Model	Method	$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 3$	$\epsilon = 4$	Non-private
Black-Box/ Shadow	DeepSeek-R1-1.5B	CoLOR-DP	0.5021	0.5031	0.5059	0.5061	0.5801
		DP-SGD	0.5064	0.5107	0.5131	0.5142	
	Llama-3.2-1B	CoLOR-DP	0.5156	0.5217	0.5256	0.5274	0.6012
		DP-SGD	0.5167	0.5238	0.5258	0.5284	
White-Box/ Partial	DeepSeek-R1-1.5B	CoLOR-DP	0.5331	0.5386	0.5404	0.5431	0.6201
		DP-SGD	0.5475	0.5491	0.5493	0.5496	
	Llama-3.2-1B	CoLOR-DP	0.5601	0.5642	0.5656	0.5667	0.6501
		DP-SGD	0.5983	0.5990	0.6003	0.6056	

Table 4: Accuracy of MIAs on IMDb dataset.

Attack	Model	Method	$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 3$	$\epsilon = 4$	Non-private
Black-Box/ Shadow	Qwen3-1B	CoLOR-DP	0.5058	0.5125	0.5157	0.5167	0.6019
		DP-SGD	0.5158	0.5187	0.5216	0.5251	
	Gemma3-1B	CoLOR-DP	0.5064	0.5070	0.5111	0.5144	0.6111
		DP-SGD	0.5117	0.5147	0.5183	0.5205	
White-Box/ Partial	Qwen3-1B	CoLOR-DP	0.5233	0.5311	0.5374	0.5401	0.6405
		DP-SGD	0.5367	0.5415	0.5455	0.5525	
	Gemma3-1B	CoLOR-DP	0.5144	0.5207	0.5261	0.5316	0.6431
		DP-SGD	0.5234	0.5296	0.5325	0.5345	

($\epsilon = 1$). With relaxed budgets ($\epsilon = 5$), the performance gap narrows, aligning with the theoretical prediction that noise impact diminishes as privacy constraints loosen. These dynamics confirm that CoLOR-DP’s structured noise injection preserves the gradient signal more effectively than isotropic perturbation.

4.2.4 Practical Recommendations for Web Deployments. CoLOR-DP provides maximum benefits when: (1) operating under tight privacy budgets ($\epsilon \leq 3$), where variance reduction yields 1.5-3% accuracy improvements over DP-SGD; (2) fine-tuning smaller models (1-3B parameters), where structured noise injection achieves 2-3% gains versus 0.5-1% on 7B models; (3) using small batch sizes (8-16 samples) typical of GPU-constrained deployments. Web services meeting these criteria gain the most from CoLOR-DP. For relaxed scenarios ($\epsilon \geq 5$, larger models), gains narrow to 0.5-1% but remain consistent. Supplementary analysis appears in Appendix C.3.

5 Related Work

Self-hosted LLM deployment requires training models on modest hardware, seamless privacy integration into existing workflows, and acceptable utility under strict budgets. We review prior work accordingly, and find no existing method addresses all problems.

Scalability to Billion-Parameter Models. Standard DP-SGD [1] clips per-sample gradients and injects Gaussian noise but scales poorly to large models, prompting dimension reduction strategies. Projection-based methods like DP-SGD-JL and D2P2-SGD [6] apply random low-dimensional projections before noise injection, achieving tighter error bounds and improved convergence. Low-rank reparameterization approaches such as RGP [31] decompose weight matrices as $W = LR^T$ and perturb the smaller factors, reducing effective gradient dimension. Memory-efficient mechanisms like DP-GRAPE [14] use Gaussian projections during backpropagation to cut memory by 60-70%, while tools like Opacus [29] enable efficient per-sample gradient computation and automatic clipping. These advances improve scalability but were designed for full-parameter training with moderate batch sizes (256+). In

web deployments with batches of 8-16, projection overhead compounds and isotropic noise impact increases. Moreover, they do not exploit the inherent low-rank structure of LoRA updates, leaving substantial room for improvement on capacity limited LLMs.

Integration with PEFT. PEFT methods align with web deployment by reducing trainable parameters and memory footprint. DP-LoRA variants [9, 30] apply DP-SGD only to adapter parameters while freezing the backbone, achieving higher accuracy than full fine-tuning with lower memory overhead. Li *et al.* [9] combine DP-Adam with ghost clipping for LoRA modules, reaching utility comparable to non-private training. Dynamic rank methods like DP-DyLoRA [28] adapt LoRA rank during training, maintaining <2% accuracy drop under strict privacy budgets in federated settings. FFA-LoRA [21] optimizes communication costs and gradient aggregation for federated scenarios. These methods reduce resource usage but inject isotropic noise without exploiting gradient structure. Moreover, dynamic approaches break CI/CD reproducibility. And all methods perturb A and B independently, ignoring the coupling that amplifies noise in BA .

Structural Opportunities in LoRA-DP. Existing projection methods treat all directions uniformly, ignoring LoRA’s low-rank gradient structure. Meanwhile, DP-PEFT methods perturb A and B independently, amplifying noise in their product BA . No prior work addresses both issues jointly. CoLOR-DP fills this gap by projecting onto rank- r' dominant subspaces and injecting negatively correlated noise, while preserving (ϵ, δ) -DP.

6 Conclusion

The fundamental mismatch between DP-SGD’s isotropic noise and LoRA’s low-rank structure forces developers to make a trade-off between privacy utility, often pushing them toward costly third-party APIs and abandoning local deployment. In order to address these issues, we presented the CoLOR-DP, which is tailored for LoRA fine-tuning in resource-constrained web deployments. By exploiting LoRA’s conjugate structure through low-rank gradient projection and correlated noise injection, CoLOR-DP reduces effective noise variance by a factor of $\frac{r'}{r}(1 - \rho)$ while maintaining rigorous (ϵ, δ) -DP guarantees. Experiments across multiple LLMs and benchmark tasks demonstrate consistent accuracy improvements over DP-SGD under identical privacy budgets, with empirical privacy validated via MIA. Our work demonstrates that privacy and utility are not mutually exclusive in self-hosted web services. By respecting the geometric structure of PEFT, CoLOR-DP enables developers to meet regulatory requirements without sacrificing the intelligence and responsiveness. Future work could extend this approach to other PEFT methods (e.g., adapters, prefix tuning) and explore federated settings via multiple web services.

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A Additional Preliminaries

A.1 LoRA Gradient Structure

The gradient structure in LoRA follows directly from the low-rank decomposition. Let L denote the loss and $G = \nabla_W L$ the gradient with respect to the effective weight $W = W_0 + \frac{\alpha}{r} BA$. Applying the chain rule yields:

Definition A.1 (Gradient Flow in LoRA). The gradients for LoRA parameters are

$$\nabla_A L = \frac{\alpha}{r} B^\top G, \quad \nabla_B L = \frac{\alpha}{r} G A^\top. \quad (19)$$

These gradients lie in a subspace of dimension $m = r(d_{\text{in}} + d_{\text{out}})$, significantly smaller than the full parameter space $d_{\text{in}} d_{\text{out}}$ yet still substantial for typical ranks ($r \in [8, 64]$). This structure becomes critical when injecting privacy-preserving noise, as discussed in Section 2.2.

A.2 Rényi Differential Privacy

Rényi Differential Privacy (RDP) provides tighter privacy accounting for iterative algorithms compared to basic composition theorems.

Definition A.2 (Rényi Differential Privacy). A mechanism $\mathcal{M}$ satisfies $(\alpha, \epsilon_\alpha)$ -RDP if for every pair of adjacent datasets D and D' :

$$D_\alpha(\mathcal{M}(D) \parallel \mathcal{M}(D')) = \frac{1}{\alpha - 1} \ln \mathbb{E}_{x \sim \mathcal{M}(D')} \left[\left(\frac{\mathcal{M}(D)(x)}{\mathcal{M}(D')(x)} \right)^\alpha \right] \leq \epsilon_\alpha. \quad (20)$$

For a query with ℓ_2 -sensitivity C , the Gaussian mechanism $\mathcal{M}_G(D) = f(D) + \mathcal{N}(0, \sigma^2 C^2 I)$ satisfies RDP with parameter:

$$\epsilon_\alpha = \frac{\alpha}{2\sigma^2}. \quad (21)$$

RDP composes additively over iterations, and converts to (ϵ, δ) -DP via:

$$\epsilon = \epsilon_\alpha + \frac{\ln(1/\delta)}{\alpha - 1}. \quad (22)$$

This closed-form accounting is efficient for tracking cumulative privacy across thousands of training steps [13], which we leverage in our privacy analysis (Section 3.4).

B Technical Details for CoLoR-DP

This appendix provides proofs for the variance analysis (Section B.1) and privacy guarantees (Section B.2) of CoLoR-DP.

B.1 Variance Analysis

B.1.1 Proof of Lemma 3.5.

PROOF. Let $z_A, z_B \sim \mathcal{N}(0, \sigma^2 C'^2 I_{r'})$ be independent. The noisy update is

$$\Delta W = (B + z_B)(A + z_A)^\top - BA = Bz_A^\top + z_B A^\top + z_B z_A^\top. \quad (23)$$

The Frobenius norm squared expands to

$$\|\Delta W\|_F^2 = \|Bz_A^\top\|_F^2 + \|z_B A^\top\|_F^2 + \|z_B z_A^\top\|_F^2 + \chi, \quad (24)$$

where χ is the cross terms. We have $\|Bz_A^\top\|_F^2 = \|B\|_F^2 \|z_A\|_2^2$ and $\|z_B A^\top\|_F^2 = \|A\|_F^2 \|z_B\|_2^2$. Since $\mathbb{E}[\|z_A\|_2^2] = \mathbb{E}[\|z_B\|_2^2] = r' \sigma^2 C'^2$, and cross terms χ vanish by independence:

$$\text{Var}[\|\Delta W\|_F^2] = \sigma^2 C'^2 (\|A\|_F^2 + \|B\|_F^2) + r' \sigma^4 C'^4. \quad (25)$$

The quadratic term $\|z_B z_A^\top\|_F^2 = O(\sigma^4)$ is omitted as it is two orders smaller. $\square$

B.1.2 Proof of Corollary 3.8.

PROOF. With conjugate noise, $z_A = \sigma C' N$ and $z_B = -\rho \sigma C' N + \sqrt{1 - \rho^2} \sigma C' M$ where $N, M \sim \mathcal{N}(0, I_{r'})$ are independent. The update becomes

$$\Delta W \approx \sigma C' (BN^\top - \rho NA^\top + \sqrt{1 - \rho^2} MA^\top). \quad (26)$$

Taking expectations and using the independence of N and M :

$$\mathbb{E}[\|\Delta W\|_F^2] = \sigma^2 C'^2 r' (\|A\|_F^2 + \|B\|_F^2). \quad (27)$$

The correlation structure reduces variance by a factor $(1 - \rho)$:

$$\text{Var}[\|\Delta W\|_F^2] = \sigma^2 C'^2 (\|A\|_F^2 + \|B\|_F^2) + (1 - \rho) r' \sigma^4 C'^4. \quad (28)$$

Substituting $C'^2 = \kappa(r') C_0^2 \approx (r'/r) C_0^2$ yields (14). $\square$

B.2 Privacy Analysis

B.2.1 Preliminary Results. We recall standard RDP results used in our analysis.

LEMMA B.1 (GAUSSIAN RDP [13]). For Gaussian distributions $\mathcal{N}(\mu, \Sigma)$ and $\mathcal{N}(\mu', \Sigma)$ with mean difference $\Delta = \mu - \mu'$:

$$D_\alpha(\mathcal{N}(\mu, \Sigma) \parallel \mathcal{N}(\mu', \Sigma)) = \frac{\alpha \|\Delta\|_2^2}{2\lambda_{\min}(\Sigma)}. \quad (29)$$

LEMMA B.2 (RDP TO DP CONVERSION [13]). $(\alpha, \epsilon_\alpha)$ -RDP implies (ϵ, δ) -DP with

$$\epsilon = \min_{\alpha > 1} \{\epsilon_\alpha + \log(1/\delta)/(\alpha - 1)\}. \quad (30)$$

LEMMA B.3 (SUBSAMPLED RDP [24]). Under Poisson subsampling with rate q , $(\alpha, \epsilon_\alpha^{\text{step}})$ -RDP becomes $(\alpha, \epsilon_\alpha^{\text{amp}})$ -RDP with

$$\epsilon_\alpha^{\text{amp}} = \frac{1}{\alpha - 1} \log(1 + q^2 \alpha (\alpha - 1) e^{(\alpha - 1) \epsilon_\alpha^{\text{step}}}). \quad (31)$$

B.2.2 Proof of Theorem 3.10.

PROOF. For adjacent datasets D, D' , the sensitivity is $\|\mu - \mu'\|_2 \leq C'/B$ by Corollary 3.9. The conjugate noise covariance is

$$\Sigma_\rho/B^2 = \frac{\sigma^2 C'^2}{B^2} \begin{bmatrix} I_{r'} & -\rho I_{r'} \\ -\rho I_{r'} & I_{r'} \end{bmatrix}. \quad (32)$$

The minimum eigenvalue is $\lambda_{\min}(\Sigma_\rho/B^2) = \sigma^2 C'^2 (1 - \rho)/B^2$. By Lemma B.1:

$$\epsilon_\alpha^{\text{step}} = \frac{\alpha (C'/B)^2}{2 \cdot \sigma^2 C'^2 (1 - \rho)/B^2} = \frac{\alpha}{2\sigma^2 (1 - \rho)}. \quad (33)$$

$\square$

B.2.3 Derivation of Corollary 3.11. By Lemma B.3, each iteration satisfies $(\alpha, \epsilon_\alpha^{\text{amp}})$ -RDP. Over T iterations, RDP composes linearly:

$$\epsilon_\alpha^{\text{total}} = T \cdot \epsilon_\alpha^{\text{amp}}. \quad (34)$$

Converting to (ϵ, δ) -DP via Lemma B.2 yields Corollary 3.11. In practice, we use Opacus RDP accountant to compute the optimal α for given (σ, q, T, δ) .

B.3 Basis-Update Sensitivity Bound

To account for the periodic updates of the projection basis U , we provide a spectral sensitivity bound using a Davis–Kahan perturbation argument.

LEMMA B.4 (SPECTRAL SENSITIVITY OF THE BASIS UPDATE). Let the per-sample LoRA gradient vectors be clipped as $\|g_i\|_2 \leq C_0$, and let the batch covariance be $\Sigma_g = \frac{1}{B} \sum_{i=1}^B g_i g_i^\top \in \mathbb{R}^{m \times m}$. Let $U \in \mathbb{R}^{m \times r'}$ contain the top- r' eigenvectors of Σ_g with spectral gap $\gamma := \lambda_{r'}(\Sigma_g) - \lambda_{r'+1}(\Sigma_g) > 0$, and let $P := UU^\top$ be the associated projector. For an adjacent batch (differing in one example) with covariance Σ'_g , define U', P' analogously. Then the projector perturbation is bounded by

$$\|P' - P\|_2 \leq \frac{2 \|\Sigma'_g - \Sigma_g\|_2}{\gamma} \leq \frac{2C_0^2}{B\gamma}.$$

PROOF SKETCH. Adjacency changes at most one outer product in Σ_g , hence $\|\Sigma'_g - \Sigma_g\|_2 \leq \frac{1}{B} (\|g_i g_i^\top\|_2 + \|g'_i g_i^\top\|_2) \leq \frac{1}{B} (\|g_i\|_2^2 + \|g'_i\|_2^2) \leq \frac{2C_0^2}{B}$. By a Davis–Kahan $\sin \Theta$ perturbation bound for invariant subspaces, the distance between spectral projectors onto the top- r'

subspaces satisfies $\|P' - P\|_2 \leq \frac{2\|\Sigma'_g - \Sigma_g\|_2}{\gamma}$. Combining yields the stated bound. $\square$

COROLLARY B.5 (PROJECTED-GRADIENT SENSITIVITY). *For any clipped vector g with $\|g\|_2 \leq C_0$,*

$$\|(P' - P)g\|_2 \leq \|P' - P\|_2 \|g\|_2 \leq \frac{2C_0^3}{B\gamma}.$$

In particular, the additional contribution to the per-iteration ℓ_2 -sensitivity from updating U can be conservatively accounted for via this bound under standard RDP composition.

This result ensures that each update of U can be safely composed under standard RDP accounting, as referenced in Remark 3.1.

B.4 Implementation Details

This section provides the detailed privacy accounting procedure for CoLoR-DP described in Section 3.5.

B.4.1 Privacy Accounting with Opacus. We use the Opacus RDP interface to determine the minimal noise multiplier σ^* satisfying a given privacy budget (ϵ, δ) . This approach uses closed-form Rényi composition and enables efficient binary search over noise levels.

The accounting procedure takes as input the dataset size N , batch size B , number of epochs E , privacy budget (ϵ, δ) , and search tolerance η . These parameters are converted to a Poisson sampling rate $q = B/N$ and total iteration count $T = E\lceil N/B \rceil$. Algorithm 2 then performs a binary search to find σ^* by evaluating the RDP bound $\epsilon_\alpha^{\text{tot}}(q, T, \sigma)$ over a grid of Rényi orders $\alpha \in \{2, 3, \dots, 256\}$.

Algorithm 2: Noise Multiplier Search for CoLoR-DP

Input: Privacy budget (ϵ, δ) , dataset size N , batch size B , epochs E
Output: Minimal noise multiplier σ^*

```

1  $q \leftarrow B/N$ ; // Subsampling rate
2  $T \leftarrow E\lceil N/B \rceil$ ; // Total iterations
3 while Binary search not converged do
4   Select candidate  $\sigma$ ;
5   Compute  $\epsilon_{\text{actual}}$  using RDP accountant with  $(q, T, \sigma)$ ;
6   if  $\epsilon_{\text{actual}} \leq \epsilon$  then
7     Update upper bound on  $\sigma$ ;
8   end
9   else
10    Update lower bound on  $\sigma$ ;
11  end
12 end
13 return  $\sigma^*$ ; // Minimal  $\sigma$  satisfying privacy budget
```

The search terminates when $|\epsilon_\alpha^{\text{tot}} - \epsilon| \leq \eta$ for all orders in the grid. After obtaining σ^* , a verification step recomputes the actual privacy cost ϵ_{spent} using a fresh accountant instance. Training proceeds only if $\epsilon_{\text{spent}} \leq \epsilon$, ensuring the privacy guarantee is satisfied. The entire search process typically requires fewer than 30 RDP evaluations.

Once σ^* is determined, no additional accounting overhead occurs during training since both the noise level and iteration count remain fixed. An important consideration is that σ^* becomes invalid if the batch size or epoch count changes after the initial search. In such cases, the noise multiplier must be recomputed to maintain

the privacy guarantee. This constraint motivates fixing all training hyperparameters before running the privacy accounting procedure.

C Additional Experiments

This section presents supplementary experimental setup and hyperparameter configuration that complement in Section 4.

C.1 Experimental Setup

Experiments run on Ubuntu 22.04 with Python 3.10, CUDA 11.8, and four NVIDIA RTX 4090 GPUs, using the LLaMA-Factory framework⁴ for LoRA integration and Opacus for RDP accounting. We compare three methods: **Non-Private**, **DP-SGD**, and **CoLoR-DP**. All methods use identical LoRA configurations to ensure fair comparison. Beyond utility metrics, we validate empirical privacy guarantees using two SOTA MIAs: **Black-Box/Shadow** attack [18], which trains shadow models to mimic the target without access to model internals, and **White-Box/Partial** attack [15], which exploits gradient information or partial model access. MIA results quantify resistance to real-world privacy threats under different adversarial capabilities.

C.2 Hyperparameter Configuration

Based on extensive experiments with LoRA ranks $r \in \{8, 16, 32, 64\}$ across benchmark tasks, we provide the following guidelines.

Projection rank r' : Set $r' \in [0.5r, 0.75r]$ as a starting point, balancing information retention and noise reduction. For common ranks $r \in [8, 64]$, we recommend $r' \in [4, 48]$.

Conjugacy coefficient ρ : Use $\rho = 0.9$ as default for language tasks. The effective range is $\rho \in [0.7, 0.95]$. Larger ρ reduces noise variance (Corollary 3.8) but may cause numerical instability. For small batches ($B < 8$), use $\rho \leq 0.85$.

Clipping threshold C_0 : Set $C_0 \in [0.5, 1.0]$ for text tasks and $C_0 \in [0.5, 2.0]$ for vision tasks. After projection, the effective threshold becomes $C' = \sqrt{\kappa(r')}C_0$ per Definition 3.4.

Noise multiplier σ : Computed automatically via the privacy accounting procedure before training. If the batch size B or epochs E change, σ must be recomputed to maintain the privacy guarantee.

Projection update frequency T_{proj} : Update the basis every 300 steps for large datasets ($N > 10^6$) or once per epoch for smaller datasets.

Additional details of privacy accounting procedures are provided in Appendix B.4.

C.3 Additional Experimental Results and Analysis

This section provides supplementary experimental analysis that complements the main findings in Section 4. We present task-specific insights, extended MIA evaluation, ablation studies, and practical deployment guidance.

C.3.1 Task-Specific Insights.

SST-2 and QNLI Analysis. These binary and question answering tasks show consistent patterns: smaller models (1-3B parameters)

⁴<https://github.com/hiyouga/LLaMA-Factory>

benefit most from CoLoR-DP. Error analysis reveals that improvements concentrate on examples requiring nuanced understanding, precisely where gradient signal preservation matters most. This size dependent improvement validates our hypothesis that capacity limited models suffer most from isotropic noise and benefit most from structured perturbation.

C.3.2 Comprehensive MIA Evaluation. Tables 3 and 4 in the main text report attack accuracy on AG News and IMDb. We provide a deeper analysis of cross-dataset patterns and attack evolution.

AGNews vs IMDb Comparison. IMDb results show slightly higher attack accuracy than AG News across all settings. At $\epsilon = 1$ with black-box attacks, IMDb yields 0.505-0.516 for CoLoR-DP versus AG News's 0.502-0.516, a small but consistent difference. We attribute this to task structure: IMDb's binary classification simplifies the attacker's decision boundary compared to AG News's four-class structure. With fewer classes, shadow models more easily learn to mimic the target's confidence patterns, improving membership inference. Additionally, IMDb's 25K training samples (versus AG News's 120K) may make individual examples more memorable, slightly elevating attack success. Nevertheless, both datasets demonstrate robust privacy protection, with attack accuracy remaining within 2-5% of random guessing under all tested budgets. This cross dataset consistency validates that CoLoR-DP's empirical privacy generalizes across data scales and task complexities typical of web applications.

Attack Evolution Across Privacy Budgets. Examining the progression from $\epsilon = 1$ to $\epsilon = 4$ reveals gradual privacy degradation as expected from DP theory, but with notably slow attack accuracy growth. On AG News black-box attacks, CoLoR-DP on DeepSeek-R1-1.5B increases from 0.5021 ($\epsilon = 1$) to 0.5061 ($\epsilon = 4$). Only a 0.4% increase despite quadrupling the privacy budget. DP-SGD shows similar stability (0.5064 $\rightarrow$ 0.5142, a 0.8 point increase). White-box attacks exhibit slightly faster degradation: CoLoR-DP increases from 0.5331 to 0.5431 (1.0 point), still remaining far below non-private baselines (0.62). This slow attack accuracy growth suggests a threshold effect: below $\epsilon \approx 3$, privacy protection remains robust, while above $\epsilon = 5$, protection may degrade more rapidly (though we do not test higher budgets). For web services, this implies that operating in the $\epsilon \in [1, 3]$ range (where CoLoR-DP also provides maximum utility gains) offers strong practical privacy with minimal risk of successful attacks.

C.3.3 Ablation Studies. We conduct ablation studies to validate hyperparameter choices and assess sensitivity to key design decisions. All ablations use DeepSeek-R1-1.5B on SST-2 at $\epsilon = 1$ unless otherwise specified.

Projection Rank r' Sensitivity. We vary projection rank $r' \in \{0.25r, 0.5r, 0.75r, r\}$ (where $r = 16$ is the LoRA rank, so $r' \in \{4, 8, 12, 16\}$). Performance peaks at $r' = 8$ (50% of r) with 91.86% accuracy, slightly outperforming $r' = 12$ (91.18%) and substantially exceeding $r' = 4$ (90.45%). Setting $r' = r$ (no projection) reverts to standard DP-SGD performance (88.88%). The results confirm our default choice of $r' = 0.5r$: lower r' over-compresses gradient information, while higher r' reduces noise reduction benefits. The relatively flat performance between $r' = 0.5r$ and $r' = 0.75r$

suggests robustness to this choice. Teams can err toward a higher r' for safety without significant penalty. Across different model sizes, the optimal r' ratio remains stable, validating our universal hyperparameter strategy.

Conjugacy Coefficient ρ Impact. Varying $\rho \in \{0.5, 0.7, 0.9, 0.95\}$ reveals expected trends: higher ρ improves utility by reducing noise variance in the product BA. At $\rho = 0.95$ (default), accuracy reaches 91.86%. Increasing to $\rho = 0.95$ yields marginal gains (92.01%), while reducing to $\rho = 0.7$ drops performance to 89.87%, and $\rho = 0.5$ further declines to 89.34%. However, $\rho > 0.95$ risks numerical instability in the conjugate covariance matrix (eigenvalues approach zero), making sampling unreliable. We therefore recommend $\rho \in [0.85, 0.95]$ as a safe range, with $\rho = 0.95$ as a robust default. Task-specific tuning can push toward $\rho = 0.95$ for maximum utility if numerical stability is monitored, or toward $\rho = 0.85$ for challenging optimization landscapes.

Projection Update Frequency T_{proj} . Updating the projection basis every 100, 300, or 500 steps yields similar accuracy (91.76%, 91.86%, 91.70% respectively), confirming that the gradient subspace evolves slowly enough that infrequent updates suffice. However, computational overhead differs substantially: $T_{\text{proj}} = 100$ increases training time by 8% versus $T_{\text{proj}} = 300$'s 3% overhead, while $T_{\text{proj}} = 500$ saves only 0.5% additional time. We select $T_{\text{proj}} = 300$ as a practical balance. For very large datasets ($N > 10^6$), $T_{\text{proj}} = 500$ may be preferable; for small datasets ($N < 10^4$), $T_{\text{proj}} = 100$ ensures projection freshness with negligible relative overhead.

C.3.4 Practical Deployment Guide.

Computational Overhead Breakdown. Detailed profiling on the model of DeepSeek-R1-1.5B reveals that CoLoR-DP's 3% overhead comprises: projection basis update via SVD (1.8%), conjugate noise generation (0.7%), and minor bookkeeping (0.5%). The SVD cost is amortized over $T_{\text{proj}} = 300$ steps, making per-step overhead negligible. Larger models (7B parameters) show similar relative overhead, indicating good scaling properties. Memory footprint remains identical to DP-SGD: both require per-sample gradient storage ($O(Bm)$ where B is batch size and m is gradient dimension), and CoLoR-DP's additional projection matrix $U \in \mathbb{R}^{m \times r'}$ is negligible ($\sim 1\text{MB}$ for typical dimensions). GPU utilization patterns are identical, as projection and noise generation are CPU operations performed during gradient accumulation.

Integration Steps. Migrating from DP-SGD to CoLoR-DP in different platform involves: (1) Add projection basis manager class (implements SVD update every T_{proj} steps); (2) Replace Gaussian noise sampler with conjugate noise generator (samples correlated (z_A, z_B) and maps via $U_{\oplus}$); (3) Adjust Opacus RDP accountant to use modified noise covariance. Existing training configurations (learning rate, batch size, epochs) require no modification.

Best Practices and Edge Cases. For very small datasets ($N < 5000$), consider reducing T_{proj} to 100 steps to track rapidly evolving gradient statistics. For extremely tight budgets ($\epsilon < 1$), increase ρ toward 0.95 to maximize variance reduction, and verify numerical stability of conjugate covariance sampling. When migrating existing DP-SGD deployments, validate privacy accounting independently using Opacus RDP to ensure ϵ budgets align with expectations.